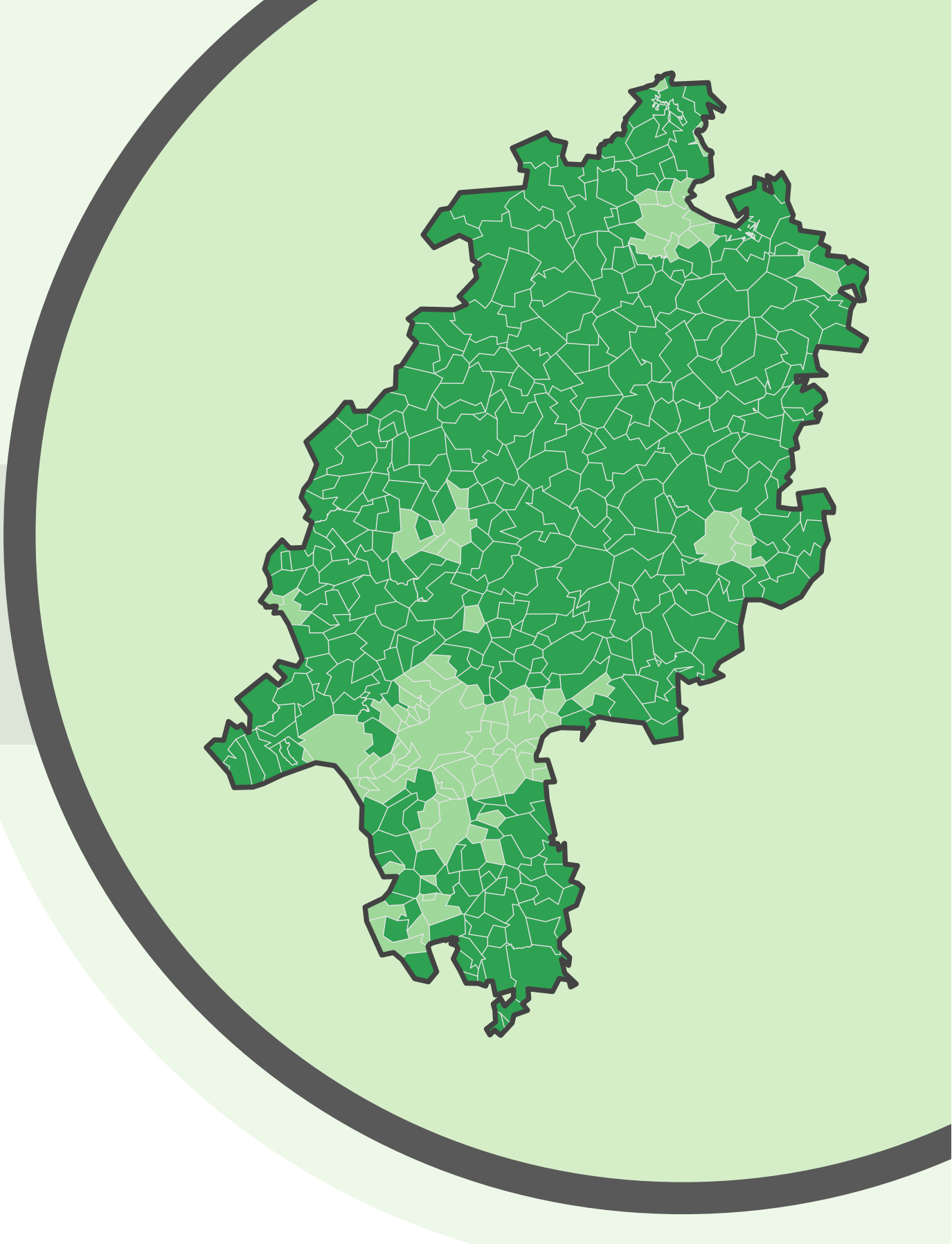


Summarizing Classed Region Maps with a Disk Choreme

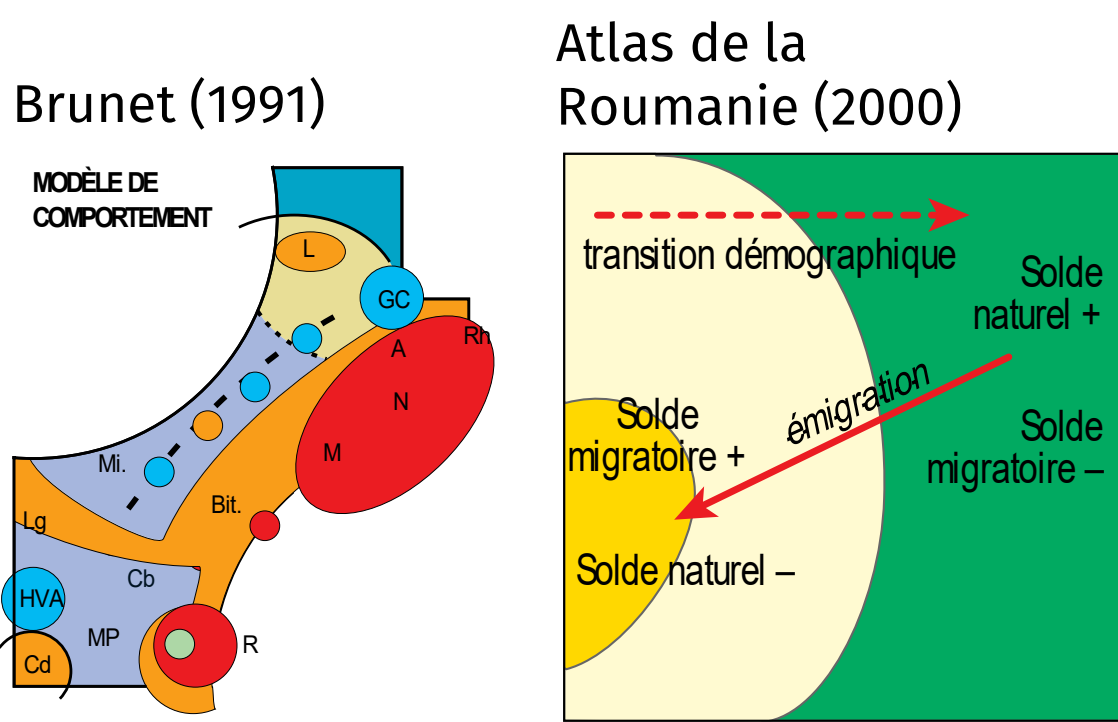
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Context

A chorematic diagram is an abstract visual summary of a geospatial situation or process. As a first step towards automation, we present a pipeline for summarizing a 2-class region map with a single disk.

Inspiration
Chorematic diagrams that use circular shapes



Problem

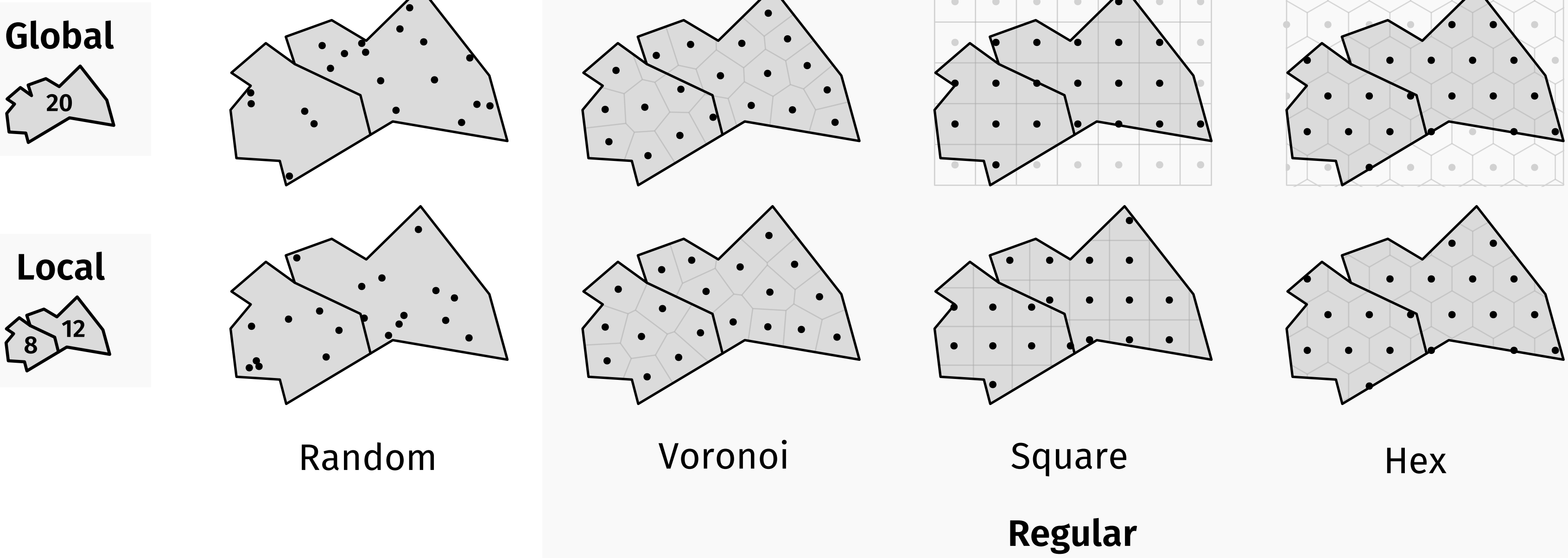
Input: two sets of pairwise-disjoint polygons S_1 and S_2
Output: disk D that best captures S_1

maximize $\alpha \cdot |S_1 \cap D| - (1 - \alpha) \cdot |S_2 \cap D|$ where $\alpha = |S_2| / (|S_1| + |S_2|)$

Solving on polygons is slow. We propose to sample points in S_1 with weight α and points in S_2 with weight $\alpha - 1$. We evaluate this approach with different sampling strategies.

Sampling

We compare four sampling techniques and two ways of applying them to the map.
Global. Ignore region boundaries; apply method to entire map at once
Local. Sample each region separately, with sample size proportional to area



Evaluation

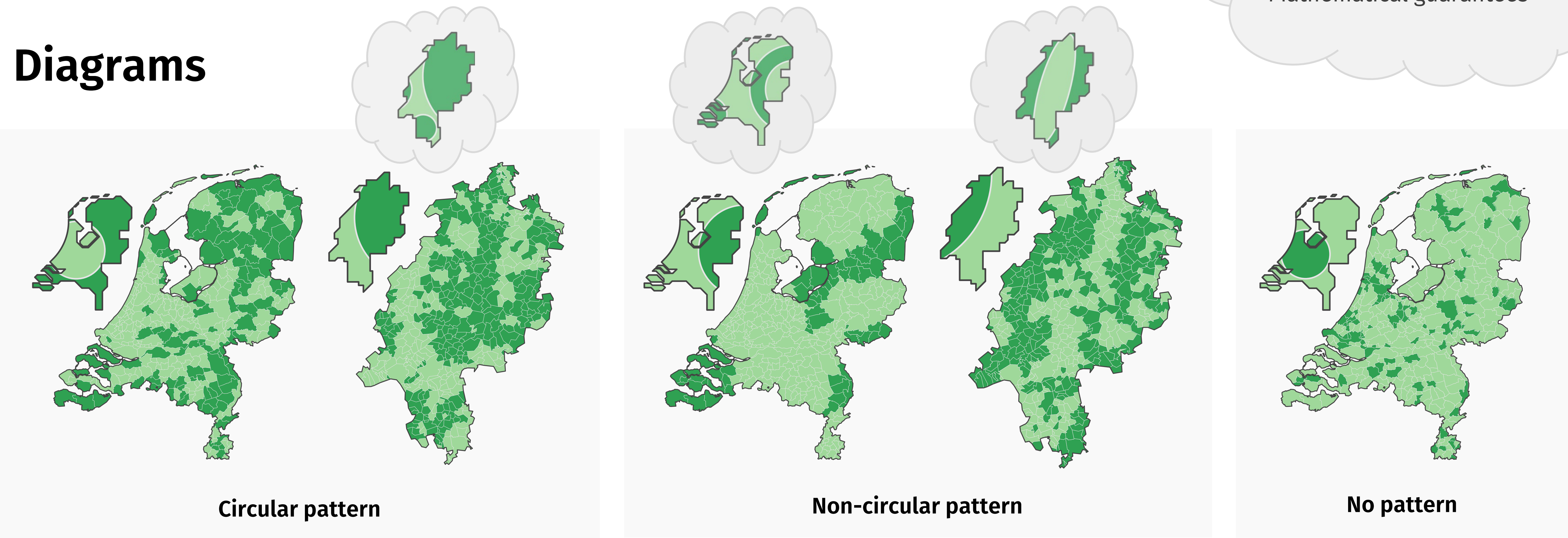
We evaluated the 8 sampling methods on 12 choropleths of The Netherlands and Hessen, Germany.

Local & Regular prevail.
Local > Global
Regular > Random

Few points suffice.
Samples of 1000 points result in high-quality disks in under a second

Future work.
More shapes
Mathematical guarantees

Diagrams

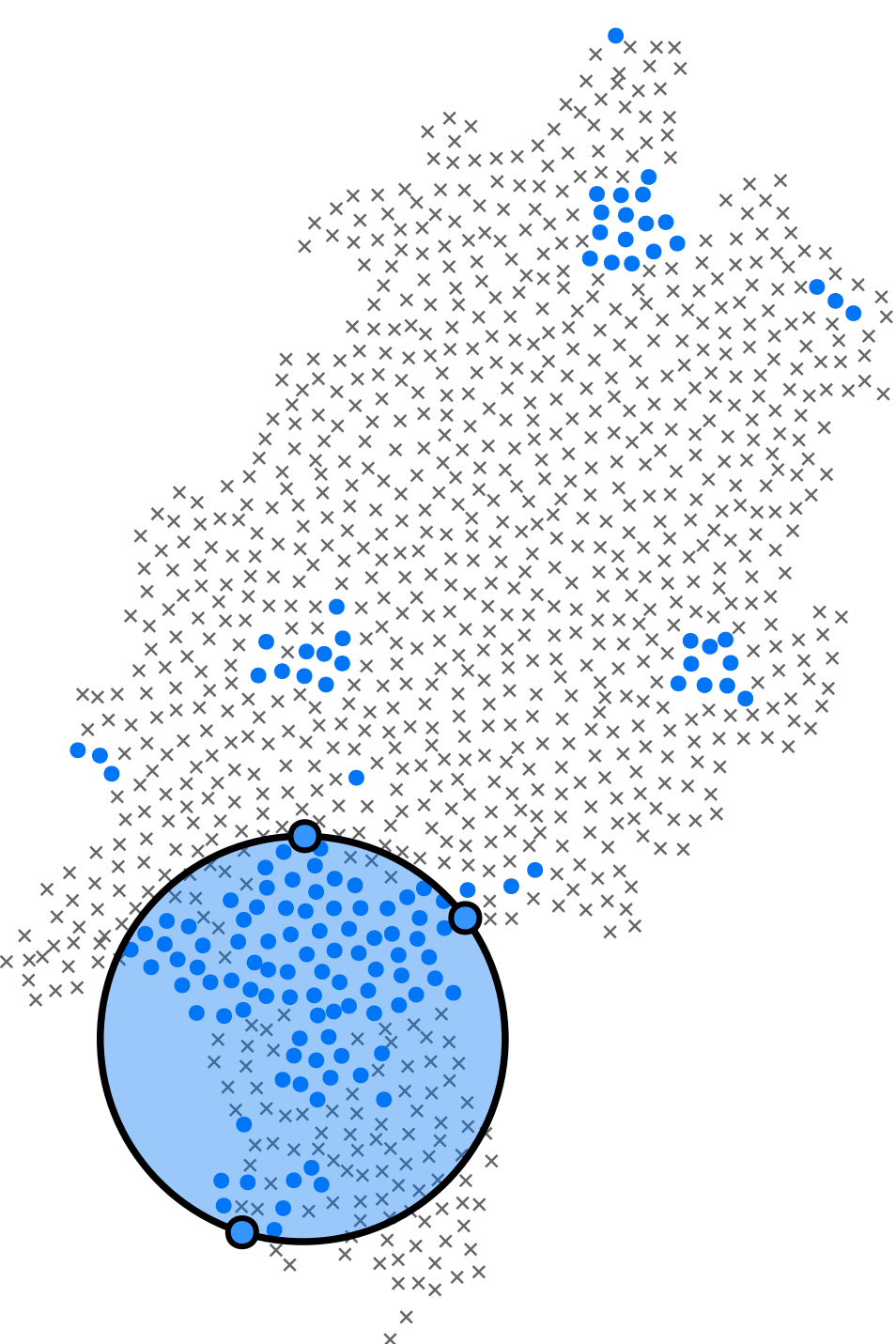


sample points

find best disk on points

Maximum-weight disk

We use the algorithm by Bereg et al. (2015) to find a disk that maximizes the total weight of covered points in $O(n^3 \log n)$ time.



Clip to schematic outline

We create an octilinear schematic outline of Hessen using the algorithm by Buchin et al. (2016).

create diagram

